Zero-Budget Lab Environment Guide

Start small, use local sample data, and add SIEM components only when ready

Design rule: the smallest useful lab first

A student does not need to install a complete SOC stack on day one. Start with a browser, a text editor, bundled sample events, and the resource directory. Add local logs and virtual machines only if the computer can handle them.

Tier	Environment	Best for	Estimated cost
Tier 0: no installation	Browser + OpenCyberPath notes + sample JSON/CSV	Foundations, reading logs, triage, metric calculations	₹0 in software for bundled files; internet/computer still needed
Tier 1: local basics	Windows Event Viewer or existing Linux logs + Python	Inspect own authorised logs, parse data, build reports	Can be ₹0 with existing OS and tools
Tier 2: local SIEM	A small local Wazuh deployment or suitable VM setup	Agents, ingestion, detection rules, dashboards	Software is open source; hardware and electricity still matter
Tier 3: advanced integration	Windows endpoint + Wazuh manager/indexer/dashboard + OpenCyberPath	End-to-end telemetry and alert investigation	Potentially ₹0 in software, but requires adequate resources and setup

Basic environment skills to learn

- File paths and permissions in Windows and Linux.
- Processes, services, users, and basic command-line navigation.
- IP address, DNS, ports, HTTP/HTTPS, and local networking.
- JSON and CSV formats; timestamps and time zones.
- Virtual machines, snapshots, networking modes, and safe lab isolation.
- Log sources, event fields, detection rules, and evidence handling.

Before installing a full SIEM

1. Check the official system requirements for the current version.
2. Check available RAM, CPU, disk space, and whether virtualisation is supported.
3. Create a VM snapshot or back up important files before changing your lab.
4. Keep intentionally vulnerable applications isolated and never expose them publicly.

Reference setup path

1. Follow the current official Wazuh Quickstart and installation requirements instead of old copied commands.
2. Start with one authorised test endpoint and one manager. Add components only as required and as hardware permits.
3. Generate a harmless test event and confirm every step: endpoint, agent/collector, manager alert, application ingestion, database record, detection, incident, and analyst note.
4. Capture evidence at each stage. A running service does not prove successful delivery of events.
5. Shut down local virtual machines and services when not in use to conserve resources.

Official docs and learning videos

- Wazuh Getting Started
- Wazuh installation guide
- Splunk free training
- Microsoft Sentinel skill-up training

COST WARNING Open-source software is not the same as zero operating cost. Full SIEM labs may need substantial resources. Avoid cloud labs or trials until you know their billing terms.

A real lab for these notes

Topic: Safe Linux, Windows CLI, and isolated practice setup. These pages connect the existing OpenCyberPath notes to specific provider-hosted exercises. The card label is intentionally generic; the exact task name and direct URL are listed here so you can open the right exercise instead of a whole catalogue.

Concept checkpoints

- Start with browser-hosted labs and synthetic files before installing virtual machines.
- Use a VM snapshot before experimenting with system settings.
- Keep intentionally vulnerable apps bound to localhost; do not expose them to a public interface.
- Never paste cloud credentials, personal files, or tokens into a training environment.

Bandit Level 0 *[OverTheWire]*
https://overthewire.org/wargames/bandit/bandit0.html
Access note: Free SSH-based command-line exercise. Use only the published game host and assigned account shown by the provider.

Linux Fundamentals *[TryHackMe]*
https://tryhackme.com/room/linuxfundamentalspart1
Access note: Account required; this is the specific Linux Fundamentals Part 1 room. Check its current access banner and machine availability.

ACCESS CHECK Open the exact task page and verify the current login, account tier, VM launch button, free/paid terms, and any cost notes before starting. A provider may change availability or retire a challenge.

Before you click Start

- Write down the provider's target/scope and use only the instance it creates for you.
- Keep this notebook beside the exercise. Use its original course data before trying commands on a local machine.
- Capture timestamps, filenames, row/packet IDs, and a short explanation; redact credentials, tokens, and personal data.
- If a page says retired, premium, or downloadable-only, use the alternative listed in the access note or return to the matching interactive lab above.

Commands you can explain

Use the published OverTheWire host or the exact machine address assigned in a training room. Do not substitute a public target of your choice.

First download the matching sample file(s) from the existing card when a command names one. Keep the data in a practice folder and never execute strings stored in evidence.

SSH / OFFICIAL BANDIT LEVEL 0

```
ssh bandit0@bandit.labs.overthewire.org -p 2220
```

Connect only to the official game host; enter the password documented on the Level 0 page when prompted.

LINUX SHELL

```
pwd && ls -la
```

Show the current directory and list files without changing them.

LINUX SHELL

```
cat readme
```

Read the Level 0 practice file after connecting, following the exact current level instructions.

WINDOWS CMD

```
ver & whoami & cd
```

Show Windows version, current user, and current directory on your own lab VM.

POWERSHELL

```
Get-ComputerInfo | Select-Object WindowsProductName,WindowsVersion,OsArchitecture
```

Read OS details from your own Windows lab VM.

LOCAL-ONLY JUICE SHOP CONTAINER

```
docker run --rm -p 127.0.0.1:3000:3000 bkimminich/juice-shop
```

Optional local web-security exercise. The 127.0.0.1 binding keeps the practice service on your own machine; stop it with Ctrl+C.

COMMAND HABIT Check the current working directory, shell, target, and expected effect before pressing Enter. Save the exact command and the output that mattered.

Guided lab run

- 1. Choose the provider-managed terminal or VM that matches the lesson.
- 2. Read the task scope and note the exact target hostname/IP, allowed tools, and time limit before launching.
- 3. Run basic read-only orientation commands first and confirm which machine you are on.
- 4. Use a snapshot or disposable container for any exercise that changes files or settings.
- 5. Stop the lab, release cloud resources, and keep evidence limited to your own training notes.

Evidence checklist

- [ ] What exact lab instance or sample file was used?
- [ ] Which command/query and time range produced the observation?
- [ ] What is the directly observed fact, separate from your inference?
- [ ] What benign/alternative explanation could fit?
- [ ] What extra record or control would help confirm the conclusion?

Check your understanding

Q Why bind a vulnerable practice app to 127.0.0.1?

Think it through: It limits access to the local machine instead of exposing the intentionally vulnerable service to the network.

Q Should a VM be bridged to a home or college LAN to make a lab work?

Think it through: Not by default. Prefer the provider's lab network or host-only/NAT networking and follow the course setup.

Q What if a command differs from the course video?

Think it through: Check the current provider documentation, OS version, and task instructions instead of blindly repeating an old command.

NOTEBOOK CLOSE-OUT Finish with: Observation / Evidence source / Why it matters / Alternative explanation / Next check / Limitation. A careful "not enough evidence yet" is a valid professional result.